

Instruction Manual of Stepper Motor Controller
Model RE-C100

Aquatech Co., Ltd.

< Configuration of the controller operation section. >

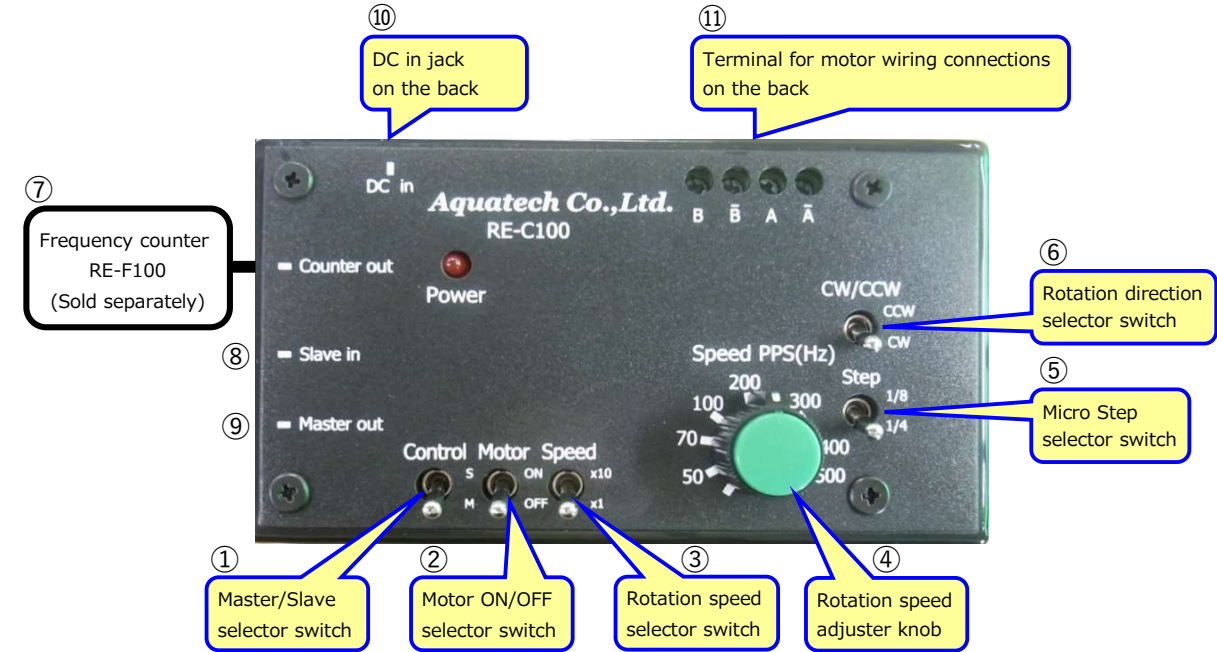
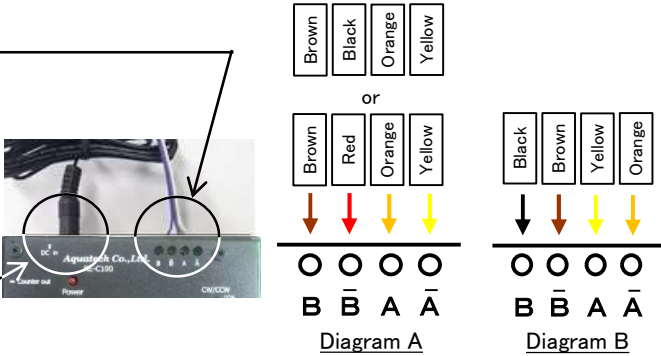


Table with 4 columns: No., Panel print, Description of the above figure, Explanation of operation/connection. It details the functions of the 11 numbered components shown in the diagram.

< How to Use the Controller >

① Connect the pump's wiring to the terminal on the controller.

Table mapping Pump series (RP-TX, RP-HX, RP-Q II B/Q III B, RP-Q II C/Q III C) to wiring color orders and connection diagrams (A or B).



② Insert the AC adapter's DC output plug into the DC-IN jack on the controller and plug the AC plug into the outlet.

③ Set the rotation speed of the motor.

Set the Step to 1/4 or 1/8 and the Speed to X1 or X10, and adjust the Speed PPS (Hz) knob. - If you are not using a frequency counter, the oscillation frequency (Hz) is obtained by multiplying the Speed PPS (Hz) dial and Speed (X1 or X10).

④ To rotate the motor : Confirm that the Control SW is on the M side, then turn the Motor SW to ON.



< Revolution conversion table >

Motor speed (pps) = Oscillation frequency (Hz) × Step[1/4 or 1/8]
Motor rotation speed (rpm) = Motor speed (pps) × 60 ÷ 20 × Reduction ratio

The relationship between the Speed PPS setting and the motor speed (pps) / motor rotation speed (rpm).

For RP-TX

Table showing Speed PPS settings (50-500 Hz) for Speed x1 and Speed x10, with columns for frequency counter, step, motor speed (pps), and motor rotation speed (rpm).

For RP-HX

Table showing Speed PPS settings (50-500 Hz) for Speed x1 and Speed x10, with columns for frequency counter, step, motor speed (pps), and motor rotation speed (rpm). Includes 'Out of step' and 'Cannot use' warnings.

For RP-Q II B/Q III B

Table showing Speed PPS settings (50-500 Hz) for Speed x1 and Speed x10, with columns for frequency counter, step, motor speed (pps), and motor rotation speed (rpm). Includes 'Out of step' and 'Cannot use' warnings.

For RP-Q II C/Q III C

Table showing Speed PPS settings (50-500 Hz) for Speed x1 and Speed x10, with columns for frequency counter, step, motor speed (pps), and motor rotation speed (rpm). Includes 'Out of step' and 'Cannot use' warnings.

< Precautions for use >

- 1. Refer to the rotation speed-discharge rate curve (included at the time of shipment) for the discharge rate.
- 2. Use the pump within the range where out of step (motor stoppage or vibration) does not occur.
- 3. TX/HX may be out of step when the motor speed exceeds 500pps. If the Step is 1/4 and the frequency counter display (oscillation frequency) is 4,400Hz or higher for QIIB/QIIBB or 4,000Hz or higher for QIIC/QIIIC, out of step may occur. Out of step is less likely to occur if the speed is gradually increased from a low speed.